THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS TWO-CHAPTER TEST SET A

Chapters 6 & 10: **Triangles, Circles** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In $\triangle ABC$, $DE \parallel BC$ with $\frac{AD}{DB} = \frac{3}{4}$ and $AE = 6$ cm. Then EC is: 1

(a) 4.5 cm **(b)** 10 cm **(c)** 6 cm **(d)** 8 cm

2. If $\triangle ABC \sim \triangle DEF$ with $AB = 4$ cm and $DE = 6$ cm, then $\frac{BC}{EF}$ is: 1

(a) $\frac{2}{3}$ **(b)** $\frac{3}{2}$ **(c)** 1 **(d)** $\frac{4}{9}$

3. Two triangles with all corresponding angles equal are similar by the criterion: 1

(a) AAA **(b)** RHS **(c)** SSS **(d)** SAS

4. A student states: “All isosceles triangles are similar.” The correct response is: 1

(a) Correct
(b) Correct only for right triangles
(c) Incorrect — their angles need not be equal
(d) Cannot be determined

5. A 2 m stick casts a 3 m shadow. At the same time a tree casts a 12 m shadow. Student A says the tree is 8 m tall; Student B says 18 m. Then: 1

(a) Student B is correct **(b)** Student A is correct
(c) Both are correct **(d)** Both are wrong

6. The length of the tangent from a point 5 cm from the centre of a circle of radius 3 cm is: 1

(a) 8 cm **(b)** 4 cm **(c)** $\sqrt{34}$ cm **(d)** 2 cm

7. Tangents TP and TQ are drawn to a circle with centre O. If $\angle PTQ = 50^\circ$, then $\angle POQ$ is: 1

(a) 100° **(b)** 40° **(c)** 50° **(d)** 130°

8. The number of tangents to a circle that are parallel to a given secant is: 1

(a) 2 **(b)** 0 **(c)** 1 **(d)** infinitely many

9. A student states: “The tangent at a point of a circle is perpendicular to every chord through that point.” The correct response is: 1

(a) Incorrect — it is perpendicular to the radius through that point
(b) Correct for the diameter only
(c) Correct
(d) Cannot be determined

10. Two concentric circles have radii 5 cm and 3 cm. For the chord of the larger circle touching the smaller one, Student A says 8 cm; Student B says 4 cm. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. In $\triangle ABC$, $DE \parallel BC$ with $AD = 2.4$ cm, $AE = 3.2$ cm and $EC = 4.8$ cm. Find DB. 2

12. $\triangle ABC \sim \triangle PQR$ with the ratio of corresponding sides 2 : 3. If $PQ = 9$ cm and $QR = 12$ cm, find AB and BC. 2

13. PA and PB are tangents from P to a circle with centre O, and $\angle APB = 70^\circ$. Find $\angle AOB$ and $\angle OAB$. 2

14. A circle touches all four sides of quadrilateral ABCD with $AB = 7$ cm, $BC = 8$ cm and $CD = 9$ cm. Find AD. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In $\triangle ABC$, $DE \parallel BC$ (D on AB, E on AC). Prove that $\frac{AD}{AB} = \frac{AE}{AC}$. 3

16. Prove that the centre of a circle touching two intersecting lines lies on the bisector of the angle formed by the lines. 3

17. In $\triangle ABC$, P and Q lie on AB and AC with $PQ \parallel BC$. If $AP = 2$ cm, $PB = 4$ cm and $BC = 9$ cm, find PQ and $\frac{AQ}{QC}$. 3

OR

The diagonals of trapezium ABCD ($AB \parallel DC$) meet at O. If $AB = 2CD$, find $OA : OC$. If $OC = 3$ cm, find AC. 3

18. From an external point P, the tangent PT is 12 cm and $PO = 13$ cm. Find the radius, and the distance from P to the nearest point of the circle on PO. 3

OR

Prove that the line joining an external point to the centre of a circle bisects the angle between the two tangents drawn from that point. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. E is a point on side CB produced of an isosceles $\triangle ABC$ with $AB = AC$. If $AD \perp BC$ and $EF \perp AC$, prove that $\triangle ABD \sim \triangle ECF$. If $AB = 10$ cm, $AD = 8$ cm and $EC = 5$ cm, find EF. 5

OR

Case study: A 4 m pole casts a 2.5 m shadow, and at the same time a building casts a 15 m shadow. 5

(i) Which similarity criterion is used?

(ii) Find the height of the building.

(iii) Later, the building's shadow is 20 m. Find the pole's shadow then.

20. **Case study:** A circular badge is inscribed in $\triangle ABC$, touching BC, CA and AB at D, E and F. $AF = 6$ cm, $BD = 4$ cm and $CE = 5$ cm. 5

(i) Find the three sides.

(ii) Find the perimeter.

(iii) Show that $AF + BD + CE$ equals the semi-perimeter.

(iv) Name the property used.

OR

Prove that the lengths of tangents drawn from an external point to a circle are equal. Hence, if tangents PA and PB from P are 10 cm long and $\angle APB = 60^\circ$, find chord AB. 5

END OF QUESTION PAPER — **Total: 40 Marks**